AKANKSHA SINGH, 39
Senior Scientist, CSIR-CIMAP
Lucknow



AZHAR

Learning from Plants

HER TAKE

"Science is not an easy path, especially for women. It needs a lot of patience and perseverance. But if you have a passion for it, it's a wonderful domain. You are constantly asking new questions and discovering new things"

Dr Akanksha Singh had once imagined herself as a doctor. But then fate brought her to botany. A PhD later, she eventually found her calling in plant-microbe interactions. Today, as a senior scientist at CSIR-CIMAP (Central Institute of Medicinal and Aromatic Plants), she is studying a microscopic world of endophytes, micro-organisms that live within plant tissues and can have a mutually beneficial relationship with their hosts.

The idea has big implications for agriculture. Unlike micro-organisms that live in the soil and are exposed, endophytes are protected within plants, making them more resilient. Some can also pass from one generation of plants to the next through seeds, raising the possibility of longer-lasting benefits. Singh's team is mapping the microbiomes of medicinal plants and some of the promising endophytes have already shown encouraging results in field trials and are being tested in crops such as wheat, rice and tomatoes.

As for her own career, after a few stutters, the turning point came in 2014 when she received the Department of Science and Technology's Women Scientist Scheme fellowship. For Singh, it became the bridge back to research. She rebuilt her career, eventually joining CIMAP.

Now, her work has received recognition on three national scientific platforms in the same year. For Singh, it's just rewards. Her aim now is to find practical ways in which the invisible microbial world inside plants can help farmers grow healthier crops, and make agriculture better equipped for a changing climate scenario. ■

—Avaneesh Mishra

WHAT MAKES HER A SHERO

A 2026 National Academy of Sciences, India, Young Scientist Award winner, Akanksha's work on plant–microbe interactions opens up new avenues in sustainable agriculture